

Memento Player: Shared Multi-Perspective Playback of Volumetrically-Captured Moments in Augmented Reality

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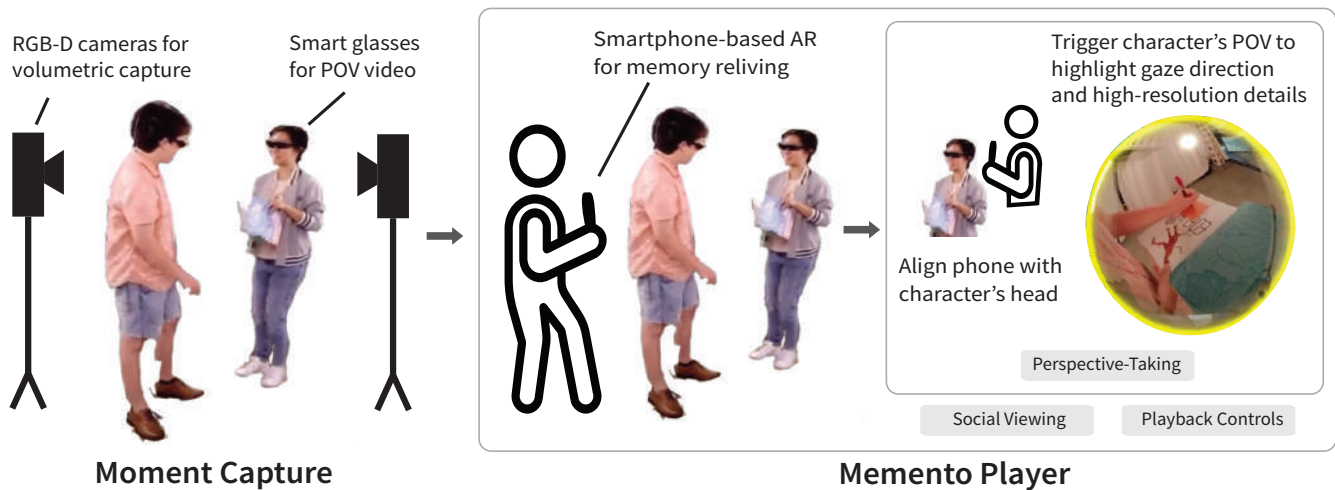


Figure 1: System overview, showing Moment Capture setup (Section 3) and Memento Player (Section 4), highlighting the immersive playback and perspective-taking features (Section 4.1). The figure on the left illustrates recording a moment with a volumetric capture (VolCap) and point-of-view (POV) videos. In the moment, a person received an autograph from a celebrity. Following the recording, a user replays the captured moment in AR using their smartphone, as shown in the figure on the right.

ABSTRACT

Capturing and reliving memories allow us to record, understand and share our past experiences. Currently, the most common approach to revisiting past moments is viewing photos and videos. These 2D media capture past events that reflect a recorder's first-person perspective. The development of technology for accurately capturing 3D content presents an opportunity for new types of memory reliving, allowing greater immersion without perspective limitations.

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In this work, we adopt 2D and 3D moment-recording techniques and build a moment-reliving experience in AR that combines both display methods. Specifically, we use AR glasses to record 2D point-of-view (POV) videos, and volumetric capture to reconstruct 3D moments in AR. We allow seamless switching between AR and POV videos to enable immersive moment reliving and viewing of high-resolution details. Users can also navigate to a specific point in time using playback controls. Control is synchronized between multiple users for shared viewing.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Mixed / augmented reality.**

KEYWORDS

augmented reality, moment reliving, volumetric capture, multi-perspective, shared and social experience

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1 INTRODUCTION

Humans have used a wide variety of media to record moments from the past, to help them re-experience past events and share them with others. People have long kept physical photo albums of cherished memories and shown home videos to allow friends and family to share moments from their life. More recently, people have begun to record videos and images with smartphones and share these captured memories with others using social media. Reliving past moments helps people develop new ways of thinking, behaving, and solving problems [41, 46], and aids with socialization [2] and identity formation [43]. These moment-reliving activities can result in positive mental health impacts [20, 48, 49, 51].

In this work, we investigate how new capture and display technologies can support re-experiencing digitally recorded events. We present Memento Player, a novel system that aims to enable a new form of shared and immersive reliving of past events using augmented reality (AR) playback of volumetrically-captured (VolCap) 3D content. While most memory-reliving experiences use 2D media, such as photos and videos, systems like Microsoft Holoportation have shown the technical feasibility of saving and replaying moments with volumetric video and re-experiencing them in 3D [39]. 3D memory re-experiencing promises a higher sense of immersion compared to traditional 2D methods [36, 57].

We build on this past work by proposing an interaction model that brings experiencing VolCap videos closer to viewing a photo album together with friends. We take advantage of the affordances of both volumetric and 2D content to allow users to step into a memory and experience it from multiple perspectives, i.e., third-person perspectives provided by volumetric video and first-person perspectives provided by point-of-view (POV) videos. Specifically, while viewing a memory, our system allows users to align their smartphones with the head of a VolCap character to trigger a 2D video showing their field of view. Combining 2D and volumetric video allows us to overcome the currently limited resolution of VolCap video and enables users to see how a moment unfolds from a specific person's perspective. Viewing memories from multiple perspectives, and attending to specific details, have been identified by prior work as important considerations for activating memories and encouraging mindful engagement [16, 50]. In addition, multiple co-located or remote users can relive moments simultaneously while maintaining an awareness of what others can see to support a social experience. Playback controls, such as pausing, rewinding, and slow motion, allow users to navigate to specific moments in time and are synchronized between users during social viewing. Figure 1 gives an overview of our system features and our moment capture setup, enabling memory reconstruction in mixed 2D+3D.

We envision these design choices for moment-reliving as important for various scenarios for capturing and re-experiencing moments. In designing our system, we were motivated primarily by

the desire to empower users to capture short but meaningful social events, such as celebrity meet-and-greets and marriage proposals. Our system could also enable reviewing and revisiting activities where people's physical capabilities change over time, such as dancing and exercising. In both cases, users can generate immersive "mementos" of essential moments they can experience alone or with others. We further detail the current system's opportunities, limitations, and ethical implications in the paper.

2 RELATED WORK

We motivate the design of our system for moment capture and reliving by contrasting it to existing systems and discussing how it relates to research on human memory and reminiscence.

2.1 Moment Capture Methods

2.1.1 Active and Passive Techniques. To support reliving, moments must first be captured *actively* or *passively*. Active moment capture, e.g., pointing a camera and taking a photo, can reveal people's personal preferences or attitudes through what they choose to capture. This is the case for many social media moments (e.g., Instagram stories), where users post photos or videos they have consciously collected and decided to share. On the other hand, passive capture, e.g., capturing sensor data [4], gaze information [31, 56], or head orientation [15] in the background, records people's behaviors without any explicit action on their part. Unlike active capture, which requires focused attention while creating a recording, passive capture implicitly documents how a person behaved and what they paid attention to during a moment. The SenseCam neck-worn camera system provides an example of how passive capture can provide a powerful memory aid [22]. Motivated by the characteristics of passive methods, in this work, we mainly use passive moment capture techniques for both 2D and 3D content to avoid distraction while people are engaged in a moment. To protect against possible invasions of privacy, while focused attention is not required during recording, users are aware of when the recording is initiated and ended with our setup.

2.1.2 2D and 3D Techniques. Most commonly used moment capture techniques, e.g., smartphones and digital cameras, capture 2D images, which provide a single point of view into a scene. However, there are an increasing number of 3D capture techniques available, such as 360 cameras, e.g., Insta360 [28], volumetric captures, e.g., Microsoft Kinect [4], and holography, e.g., Tensor Holography [45]. Volumetric capture technology allows for the recording of entire 3D scenes. One common VolCap technique uses multiple RGB-D cameras to record a scene from different viewpoints. The collected RGB information is then projected at the correct depth in 3D content creation, so a user can move freely in space while viewing the captured content. Several technology companies, such as Google, Microsoft, and Sony, have used this approach to produce various applications for, e.g., education/training [17–19] and personal memories [9, 13, 39, 44] in augmented, virtual and mixed reality (AR/VR/MR). In this work, we combine 2D and 3D moment capture techniques; volumetric capture technology is used for recreating moments in 3D to create a sense of immersion and presence, while point-of-view videos in 2D provide high-resolution details and capture information about the camera wearer's gaze direction.

2.2 Moment Replaying in Augmented/Virtual Reality

Augmented and virtual reality have become popular methods for viewing and interacting with 3D content, providing increased immersion and a sense of presence compared to traditional desktop displays [8, 21, 34, 36, 57, 58]. Although both AR and VR support immersive experiences, the bulkiness of head-mounted displays and the fact that they obscure a user's normal vision limit one's ability to interact with and maintain social awareness of people around them in the physical world, especially for VR users. To enable social awareness while re-experiencing past moments (Section 2.3.4), we chose smartphone-based AR as a platform. AR glasses, such as Snap Spectacles [27], were an alternative option, but were not used in our system for moment reliving due to their shorter battery life and smaller field-of-view compared to smartphones. With continued advancements in AR display technology, the current pipeline could be extended to AR glasses to gain improved immersion.

2.3 Moment Reliving

2.3.1 Values of Moment Reliving. Moment-reliving technology can be helpful to support people in reminiscing about meaningful episodes from their past. Prior work has shown that recalling positive events can safeguard people against depression, anxiety, and stress [20, 48, 49, 51]. For people with dementia, positive moment re-experiencing can help their caregivers understand their past, thereby improving their quality of life [47, 54].

In particular, Askelund and colleagues [2] emphasized that recalling detailed positive moments specific in time and location is more beneficial than general recall. Motivated by related psychological theories, Gldenpfennig and colleagues [16] built several prototypes that explored how focusing on specific visual details can promote more mindful engagement while reminiscing. These findings show the value of highlighting details in a moment-reliving experience.

Inspired by this line of related work, we propose that moment reliving should not be limited to passively watching a static video of a recorded event; instead, it should allow participants to recreate an experience and encourage them to mindfully engage with and explore the moment, to discover unnoticed or serendipitous details.

2.3.2 Perspectives of Moment Reliving. Viewing a moment as seen through one's own eyes is *first-person perspective* moment reliving [53]. Although we typically think of memories as matching the perspective we experienced them in [30], research suggests that memories can shift over time to match an observer's point of view, leading people to revisit a past moment and see themselves from a *third-person perspective* [5, 7]. Reviewing this line of research, St. Jacques [50] summarized how reliving a moment in a third-person perspective can reshape how a moment is remembered over time.

Inspired by this research on shifting perspectives during past moment retrieval, we designed our system to enable users to take on the perspectives of themselves, another person, or an invisible observer during a moment-reliving experience. Such design not only offers first- and third-person perspectives, but a *second-person perspective* that depicts the experience from the point of view of others present during the moment. For example, while

re-experiencing a birthday party, a user can look from their own eyes or see themselves from their parents' viewpoints. Reviewing memories by switching between multiple points of view allows users to discover what they could not see, hear, or feel originally.

2.3.3 Speed of Moment Playback. Human visual perception has temporal limitations such that we cannot observe events that happen too quickly, especially when multiple events happen in parallel. Knierim and colleagues [33] introduced a proof-of-concept system to study how slow motion could affect human perception by conducting a visual diagnosis evaluation in AR. Their user participants reported improved perception and confidence by slowing events down. Motivated by this work, we incorporate multiple controls to allow users to manipulate playback speed and investigate specific segments of a captured moment. While video-style playback control does not necessarily match how memories are represented or re-experienced in the human brain, we prioritize the ability to review visual details (e.g., by pausing) over naturalness.

2.3.4 Sharing & Reliving Moments Socially. Re-experiencing happy moments, especially those involving social interaction, benefits people [2]. When reliving a moment, people often share it with friends and family, inviting them to re-experience that moment jointly. Sharing a moment and reliving it with others can be co-located or remote. A co-located experience allows participants to re-experience a moment in a shared physical space - like telling childhood anecdotes to a friend while sitting in your old bedroom. Participants can maintain awareness of each other and engage in physical social interaction. In addition to synchronous co-located moment reliving, asynchronous co-located experience has been studied by Memon and colleagues [38] and Zhang and colleagues [60] where location information is used as a trigger for friends to relive a shared past moment when they check in to the same location or similar context. On the other hand, a remote moment-reliving experience connects participants virtually (e.g., collectively reflecting on content surfaced by Facebook Memories). While this may allow more people to view a moment from the past, a remote co-reliving experience is less rich than co-located sharing due to the lack of in-person interaction. Due to the value of social reminiscence, we implemented collaborative features to support both co-located and remote moment reliving. In both cases, users can maintain awareness of others' movements with virtual avatars.

3 MOMENT CAPTURE

To capture moments, we use VolCap to record videos of humans interacting with each other and with objects. Specifically, we use nine Microsoft Kinect [4] sensors that capture color and depth information. Eight cameras are positioned evenly at a constant height in a circle surrounding the VolCap stage, and a downward-facing camera is positioned above the center of the stage. Cameras are calibrated and aligned using the Depthkit software package [11]. We decided to use nine sensors to ensure high quality for both recording and reconstruction. As shown by a recent moment reliving platform named Wist [59], missing information (such as a subject's back) occurs in moments captured using a single-perspective video. Our setup captures a moment from 360 degrees to eliminate such

significant gaps in the reconstructed content without requiring advanced techniques, such as monocular reconstruction, which may hallucinate details and still result in missing body parts [1]. Our setup helps envision a higher fidelity moment capture experience that could become more available and common in the future, with the development of more accessible computer vision techniques and cheaper, more portable hardware, such as mini drones and depth cameras. Similar to Google Starline [13], where a sophisticated setup is required to demonstrate a future for live 3D communication, our system shows the possibility of AR moment reliving experiences that improve on those widely available today.

The stage is lit with four evenly distributed lights at four corners to avoid color imbalance and covered by white drapes to block out background noise. The stage area is approximately $8' \times 8'$, allowing for the recording of one or two person(s). Figure 1 demonstrates a VolCap example with two people being recorded, and Figure 5 in the appendix shows the VolCap stage. We clean the data collected by the RGB-D cameras by applying human and object segmentation using U2-Net [42] to remove background pixels, such as cameras and drapes. Following data processing, we reconstruct moments in AR using Snapchat Lens Studio [25].

Although VolCap supports the 3D reconstruction of recorded moments, suitable for display in AR, it remains challenging to capture high-resolution information, e.g., people’s facial expressions or text on objects. Furthermore, VolCap records a moment from a neutral third-person perspective without reflecting a human’s subjective attitudes. On the other hand, the first-person perspective from a POV video conveys how a person felt and what they paid attention to during the moment. Similarly, second-person perspectives from other participants hint at their respective preferences and focus. Considering the limitations of current VolCap technology, we propose integrating POV videos captured by AR glasses (in our case, Snap Spectacles [27]), in addition to the reconstructed VolCap content in a single AR moment-reliving experience. We choose AR glasses rather than smartphones or cameras to record first-person POV video because users do not need to explicitly think about what to capture, as is required when holding a phone. This allows users to behave more naturally, be less distracted, and maintain immersion during moment capture. For both POV and VolCap recordings, participants are notified when the recording starts and ends.

4 MEMENTO PLAYER

To support the playback of captured content, we designed the Memento Player system. We built Memento Player in Snapchat [26], a popular multimedia instant messaging app that supports mobile AR experiences through custom lenses created in its Lens Studio development environment [25]. In this section, we elaborate on the design details of our system that enable immersive viewing and perspective-taking, playback control, and shared moment-reliving.

4.1 Immersive Viewing and Perspective Taking

As a user navigates in AR, the distance and angle between the user’s viewpoint relative to the VolCap content and each sensor position is dynamically computed, so that the closest sensor’s RGB-D projection is presented to the user.

To trigger a first-person perspective, users approach and align their phones with a character in the volumetric video, causing the VolCap content to disappear and revealing a POV video element. Figure 2 shows a user approaching a character in the AR environment to trigger a POV video. We project the videos onto a sphere to create a fish-eye effect, increasing the resolution of details near the center of the field of view. Additionally, users can scale POV videos by moving their phones forward and backward, and pan by moving left and right. Users can smoothly switch from the POV videos back to the AR environment by stepping away from the reconstructed human characters.

4.2 Playback Control

Instead of passively watching moments on an infinite loop, we design multiple playback controls (for both the VolCap content and POV videos), including pausing, rewinding, fast-forward, and slow-motion. These controls enable users to focus on key points in time within a moment. For example, by pausing the moment, a user can explore a single instant from multiple angles, as shown in Figure 3, similar to Intel freeD sports replay technology [29]. By slowing down playback, events that happen too quickly to process in real-time can be reviewed in detail; by rewinding the moment, users can repeat particular highlights over and over. Captured audio is also affected by the playback controls, which results in whimsical audio effects during rewinding, fast-forwarding and slow-motion.

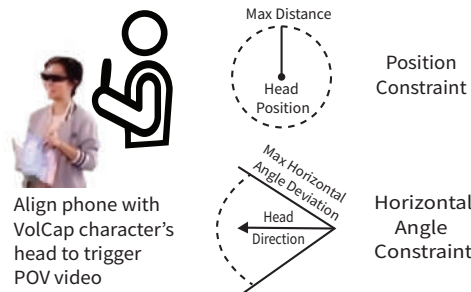


Figure 2: Overview of perspective-taking interaction. A user aligns their phone with a volumetrically-captured character’s head, within the position and angle constraints, to trigger a POV video that reveals high-resolution details with a fish-eye effect.



Figure 3: Overview of playback controls. A point in time is viewed from multiple angles after pausing the volumetric video.

4.3 Shared Moment-Reliving Experience

To allow our system to take advantage of the positive impacts brought by social reminiscence (Section 2.3.4), we develop a shared and social moment reliving feature based on Connected Lenses [24] in Lens Studio. The system allows users to invite remote or co-located friends. In both cases, the content and playback controls are synchronized between users in the same moment-reliving experience, so users can guide friends through the process of watching a moment together. In particular, the co-located AR setup enables users to engage socially by talking and observing each others' reactions to the moment (see Figure 4). Such awareness is challenging to achieve in other systems. For instance, in co-located VR, headsets would obscure facial expressions, inhibiting communication. This benefit of mobile AR serves as an important enabler for building a connected and social moment-reliving experience. In contrast, in the remote setting, the content remains synchronized, but only the smartphone avatar indicates other users' locations.

5 DISCUSSION

5.1 Opportunities

Memento Player allows us to explore a possible future for interaction with recorded moments from the past. Presently, short moments from the past can be captured volumetrically at a reasonably high degree of fidelity. In the future, people will likely have access to volumetric reconstructions of a much wider range of important past moments. In this project, we investigated what types of interaction could be added to captured moments to support users in re-experiencing them. As capture technology improves, we believe

that the design considerations identified here and addressed by our prototype features will continue to be relevant.

Our current setup is suitable for volumetric recording of one or two person(s) involved in activities of up to roughly 60 seconds. We are particularly interested in recording meaningful social events, such as meeting with a favorite celebrity to request an autograph or a marriage proposal to a romantic partner. These activities are feasible to record within our current time and space constraints and are the events users might truly want to revisit at multiple points in the future. The inclusion of social viewing features in our system aligns with the desire of users to share their memories with others (Section 2.3.4), which may be particularly strong for this type of meaningful event. We are also interested in recording activities where people's physical capabilities change over time, such as dancing or working out in place. When people relive these events, our system could support their understanding of the development of their bodies and skills over time. Replaying such recordings from multiple angles and at various speeds could also provide training materials for others to learn these activities.

By viewing the volumetric content in AR, we aim to give our users a deeper sense of immersion and presence compared to viewing photos on a conventional 2D display. We are interested in understanding whether the sensation of physically "stepping inside" a moment from the past can allow users to understand and relate to the content in new ways. Playback controls give users control over how they re-experience the moment, allowing them to browse quickly or focus on particular points in time. The perspective-taking interaction enables users to view the way events played out through the eyes of a specific individual who was present, understand what

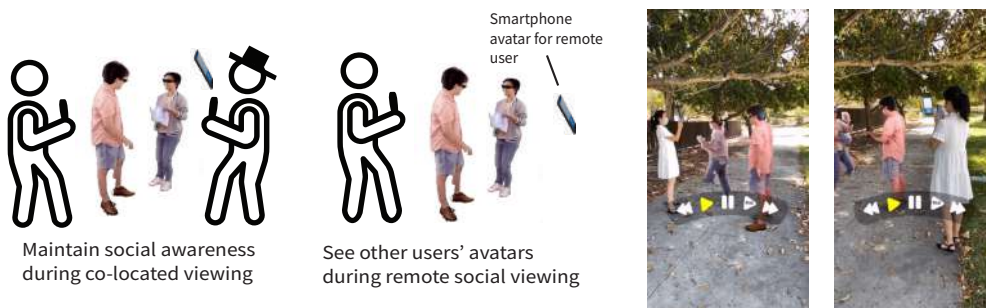


Figure 4: Overview of social moment reliving. A connected user and phone location are present during co-located viewing.

they paid attention to at the moment, and discover overlooked details. While the current inclusion of POV videos is important for overcoming VolCap quality limitations, perspective-taking itself would remain useful and meaningful even with perfect volumetric capture, because it allows users to access this subjective experience. To support multi-perspective experiences like the one offered by Memento Player, future work could investigate how to automatically extract a user's POV video from volumetrically captured content.

5.2 Limitations

Practical limitations restrict the types of content our system can capture and replay. To reach the highest quality given our current setup, we set the color resolution during data recording to 1440p. Following the recording step, we upsample the color resolution to 2880p. To process the high-resolution data, we compute masks for each frame to filter out background information, which we apply to both RGB and depth images. The approach we use to obtain the masks is a semantic segmentation neural network (Section 3). The deep learning model inference and computed mask matting take approximately 90 minutes for a 1-minute VolCap recording. Considering the time taken for data processing, the length of volumetric captures is limited to 1 minute. Recordings of over 30 seconds require an external camera in addition to the AR glasses, as Spectacles are currently limited to 30-second recordings.

Due to the time taken for data processing, though we are interested in passive capture, it is currently impractical to record long activities, and users must consciously choose to begin and end recording. It is also impractical to record activities that require movement (e.g., running) because of the static capture rig which is space-limited. In addition, we experimented with activities requiring props, such as skateboarding, during which the participants carefully performed their actions with our guidance. However, they could still lose control and accidentally hit the capture rig, requiring time-consuming re-calibration.

5.3 Ethical Implications

Our choice to expand the types of content that can be passively recorded and immersively replayed raises some ethical issues related to personal privacy. For example, people may not be aware of how they appear from all angles, such as from the back. Passively capturing the user's gaze during a moment may also reveal information they might wish to keep private about what they were paying attention to. The current setup minimizes the risk of recording information users want to keep private, since they must consciously enter the inward-looking volumetric capture rig, and are aware of when recording begins and ends. However, if immersive memory capture becomes more pervasive in the future, such issues may need to be confronted. In particular, who controls when and in what situations passive memory capture takes place is important, and recording people without their consent or knowledge is likely inappropriate. Early prototypes like Memento Player might serve as tools to explore user opinions related to privacy.

In addition to privacy, the choice to convert impermanent moments into permanent recordings raises further questions about how having access to an immersive capture might reshape a user's

memory of an event over repeated viewings. While individual and social reminiscence is already supported by existing technologies (such as digital photo albums and video galleries [3]), whether long-term immersive replaying might cause different affective responses in users should be considered.

6 FUTURE DIRECTIONS

6.1 User Evaluation

An immediate next step for future work would be to validate the system's design by evaluating it with users. This would allow us to test whether the interaction model achieves the goals we have described, whether users see value in the perspective-taking, social viewing, and playback control features compared to a static looping VolCap memory, and what additional supports (e.g., user interface hints to make the capabilities of the system discoverable) would be necessary. In addition, we could evaluate whether the fidelity and usability of the current prototype are sufficient and whether the playback controls match users' mental models for traversing memory experiences. We are also interested in exploring differences between the remote and co-located social viewing experiences to understand what additional features might bring remote social viewing closer to the experience of co-located immersive reminiscence. Inviting users could further elicit feedback on what scenarios this type of system could suit, what ethical concerns people might have, and what additional features people would like to suggest for future moment-reliving systems.

6.2 Larger Space and Higher Quality

Our current volumetric capture setup is limited by stage size and the number of VolCap sensors, which restricts the number of people we can capture and the activities they can participate in. With access to additional sensors and improved lighting conditions, we anticipate that our current pipeline could extend to a larger capture area and higher capture quality. To process VolCap data from more sensors and of longer lengths, additional computing resources, such as GPUs, would also be needed. In the future, we envision applying VolCap to public events, such as concerts, and allowing people to relive these moments in an immersive and spatial manner.

6.3 Triggers of Moment Reliving

One aspect of moment reliving we did not consider is when users might want to engage in such an experience. In this work, we capture three major entities in a moment: people, objects, and activities, which could be used as triggers to prompt users to engage in moment-reliving (e.g., sending a notification suggesting two friends open Memento Player when they are physically close, or when one of them is involved in the type of activity they previously recorded). Prior research has explored many additional triggers to initiate memory recalling, such as time [14], location [6, 12, 37], emotion [41], weather [35], touch [23, 40], smell [32, 52, 55], and taste [10]. Future work could test and evaluate how users respond to different types of triggers.

7 CONCLUSION

Memento Player demonstrates a new way of reliving past moments immersively through social and interactive AR playback of volumetrically-captured moments. To fill the gaps in existing memory re-experiencing techniques, platforms, and systems, we propose combining 2D and 3D moment captures, enabling perspective-taking, and maintaining social interaction during viewing. Through these design choices, Memento Player allows users to collaboratively explore and relive an experience rather than passively watching a captured scene. We hope this work will encourage future research into design considerations for moment-reliving systems.

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Figure 5: Volumetric capture stage from two angles.

A VOLUMETRIC CAPTURE STAGE

Figure 5 shows the layout of the Volumetric Capture setup.

B VOLUMETRIC CAPTURE QUALITY

The quality of RGB-D reconstructed VolCap content is limited by the current VolCap stage setup. As introduced in Section 3, the stage is covered by white drapes to eliminate complex background noises, but there exist wrinkles on the drapes such that the drape color is uneven. This noise can introduce challenges to the human and object segmentation model and thus cause boundary artifacts in the VolCap content. In addition, the drape color is white, causing

issues when users wear light-colored clothes due to the need for segmentation. We plan to test with traditional green or blue screens for the stage setup.

A related quality limitation caused by segmentation is using U2-Net in our pipeline. This model usually segments dark-colored regions as background, leading to holes when users wear dark clothes. This might have to do with the lack of dark-clothed humans in the U2-Net training dataset. Future work could experiment with different human and object segmentation models to improve performance, as these errors significantly reduced the quality of volumetrically-captured 3D content.